30-7-23

Physics

Ans.1 Convex Lens is used to produced real image.

Ans. 2 $Power(P) = \frac{1}{Focal\ length(f)\ in\ m}$

In case I :-

$F = 10\ cm$ given

$F = \frac{10}{100} = 0.1m$

$P = \frac{1}{0.1} = 10D$

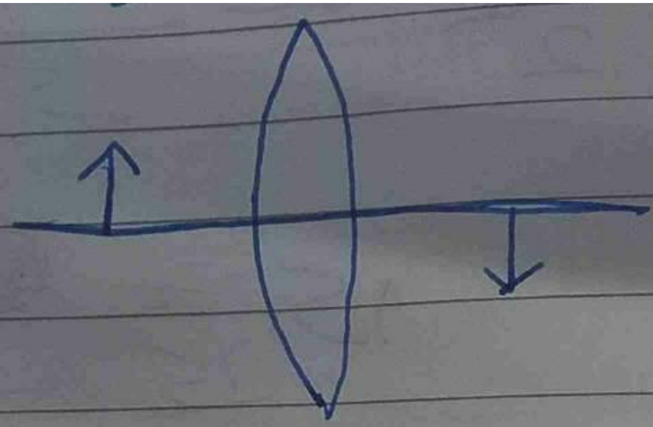
In Case II

$F = 20\ cm$

$F = \frac{20}{100} = 0.2$

$P = \frac{1}{0.2} = 5D$

Ans. 3



$U = -5 \, cm$

$V = +10 \, cm$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

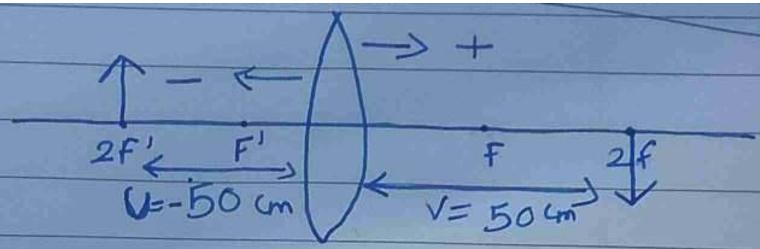
$$\frac{1}{F} = \frac{1}{10} + \frac{1}{5}$$

$$\frac{1}{F} = \frac{1+2}{10} = \frac{3}{10}$$

$$F = \frac{10}{3} = 3.33$$

$\therefore$ It is convex lens because F is +ve

Ans.4 $U = -50$ cm
$V = 50$ cm



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$$\frac{1}{F} = \frac{1}{V} - \frac{1}{U}$$

$$\frac{1}{F} = \frac{1}{50} - \left(\frac{1}{-50}\right)$$

$$\frac{1}{F} = \frac{1}{50} + \frac{1}{50}$$

$$\frac{1}{F} = \frac{2}{50}$$

$$\frac{1}{F} = \frac{1}{25}$$

$F = +25$ cm

$F = \frac{25}{100}$ m

$F = 0.25$ m

Ans. 4 $\text{Power} = \dfrac{1}{\text{Focal length}}$

$$P = \frac{1}{F}$$

$$P = \frac{1}{0.25} = +4D$$